

The LHS 1140 b catalog: what is certified and what is not

Every model of the catalog is LHS 1140 b on 500 cells, solved by the partitioned stationary route with `Well balanced: True`, from the GJ 1132 spectrum placed at this planet's orbit. The binary of record is `EXHALE_75d55d9d.x`, md5 `75d55d9d4fd0e748cd01d6e35713e34c`, and the counts below are MEASURED by `models/status.py` inventory, re-run read-only at 2026-09-22 11:55:41 KST: **143 state indexes, 92 certified**, the same totals the stored `CASE_INVENTORY.json` carries from its 2026-09-21 21:10 reading. A case counts as certified when its index names a `latest_certified` generation that no refused re-evaluation has marked stale. Section 3 and section 4 list the 94 models of the ladders and of `diffusion_check`; the remaining 49 indexes are 46 flux-closure iterates, two archival copies and one study directory.

Where each column comes from. du and $\|R\|$ are both read from the LAST (JFNK) done line of the case's run.log, so they are one pair printed by one solve and not two numbers from two places; where a case was last re-measured by an evaluation that took no step, that line is the solve preceding it, and where a case was never advanced there is none and the columns read --. The refusing rows of section 3 are read from `certification.txt` of the generation the case's index names `latest_complete`, and EVERY refusing entry of that certificate is listed, not only the first. $\log_{10} \dot{M}$ and the equivalent width are from `models/status.py`, which reads them from the case directory. The XUV column is the factor the stellar spectrum is scaled by; K_{zz} is the eddy coefficient of the base, *profile* where a photochemical column supplies $K_{zz}(p)$, and *none* where the base is well mixed. The *wind chemistry* column of every table below says what is solved ABOVE the base, atomic or molecular in the sense of section 1; what supplies the base itself is in the case name (scalar or photochem) and in the K_{zz} column.

1. Two axes: where the lower atmosphere comes from, and what the wind solves

A case name carries both axes, and they are independent. The first says what supplies the state at the base level, the second whether the wind above it carries a molecular network at all.

The lower boundary.

- scalar: the base is a few prescribed numbers at the level, read from `base.inp`: the pressure (1 microbar), the temperature (226 K), the base radius, the bound hydrogen fraction $q_{\text{H2,base}}$ and the element reservoirs. Nothing below the level is modeled.
- scalarCNO: the same scalar base with the C, N and O reservoirs of the photochemical column added at the matching level.
- photochem: the lower atmosphere is a Photochem column handed over as a PROFILE, the key Lower atmosphere profile: naming a file `lower_atmosphere_profile.dat`, which carries $T(p)$, $K_{zz}(p)$, the bound hydrogen fraction and the element reservoirs at the matching level $p_{\text{match}} = 1$ microbar. For the atomic groups the reservoir He/H is then solved by the elemental-flux closure (`src/utls/element_flux_closure.py`), which

iterates the column and the wind against each other: the iterates are the $k\langle NN \rangle$ directories, and the composition is an output of that loop rather than an input.

The wind's chemistry.

- atomic: hydrogen and helium with their ionization stages and, where the group says so, trace metals. No molecular network is solved above the base, whatever the base carries.
- molecular: EXHALE's own network, Molecular chemistry: True with Molecular carrier transport: True, solving H_2 , H_2^+ , H_3^+ and HeH^+ in the wind with the H_2 carrier transported. The molecular base it starts from is the scalar handoff or the profile, by the other axis.

So atomic_photochem... is a Photochem column under an atomic wind, and molecular_scalar... is EXHALE's own molecular chemistry above a scalar base. The two are not alternatives for the same layer: the Photochem column is the atmosphere BELOW the matching level, and EXHALE's network is the chemistry ABOVE it. A molecular_photochem case uses both, the column below and the network above.

	wind solved atomic	wind solved with EXHALE's molecular network
scalar base (base.inp)	atomic_scalar_*	molecular_scalar_*
Photochem column (profile)	atomic_photochem_*	molecular_photochem_*

The mixing suffix is a third, smaller axis: wellmixed is no element diffusion, kzz1e9 is binary H/He diffusion at that eddy coefficient, and kzzprofile takes $K_{zz}(p)$ from the column.

2. By group

group	wind chemistry	He/H	XUV	K_{zz}	diff.	du reached	cases	cert.
diffusion_check	atomic	0.55	1	0.0 to 1.0e9	on	–	4	0
photochem_x0.01_kzzprofile	atomic	9.7	0.01	profile	on	9.4E-03	1	0
photochem_x0.10_kzzprofile	atomic	9.7	0.10	profile	on	5.0E-10	1	1
photochem_x0.15_kzzprofile	atomic	9.7	0.15	profile	on	4.7E-12	1	1
photochem_x0.20_kzzprofile	atomic	9.7	0.20	profile	on	2.0E-12	1	1
photochem_x0.25_kzzprofile	atomic	9.7	0.25	profile	on	1.9E-12	1	1
photochem_x0.30_kzzprofile	atomic	9.7	0.30	profile	on	1.3E-10	1	1
photochem_x0.33_kzzprofile	atomic	9.7	0.33	profile	on	1.6E-10	1	1
scalarCNO_kzz1e9	atomic	2.13	1	1.0e9	on	1.7E-11	1	1
scalar_gj699_wellmixed	atomic	0.042 to 1000	1	none	off	1.8E-13 to 1.6E-11	6	6
scalar_kzz0	atomic	0.55 to 3.9	1	0.0	on	2.4E-11 to 1.2E-10	6	6
scalar_kzz1e10	atomic	0.55 to 1.29	1	1.0e10	on	2.0E-11 to 2.7E-11	4	4
scalar_kzz1e11	atomic	0.55 to 0.93	1	1.0e11	on	1.8E-11 to 1.9E-10	4	4
scalar_kzz1e5	atomic	2.6 to 3.93	1	1.0e5	on	2.5E-11 to 4.8E-11	5	5
scalar_kzz1e6	atomic	0.55 to 3.74	1	1.0e6	on	1.9E-11 to 6.8E-11	6	6
scalar_kzz1e7	atomic	0.55 to 3.18	1	1.0e7	on	1.6E-12 to 1.3E-10	4	4
scalar_kzz1e8	atomic	0.55 to 2.41	1	1.0e8	on	5.5E-11 to 7.9E-11	4	4
scalar_kzz1e9	atomic	0.55 to 9.7	1	1.0e9	on	9.3E-13 to 6.1E-11	7	7
scalar_wellmixed	atomic	0.083 to 1000	1	none	off	1.4E-12 to 1.2E-10	9	9
scalar_x0.015_kzz1e9	atomic	2.13	0.015	1.0e9	on	1.8E-10	1	1
scalar_x0.01_kzz1e9	atomic	2.13, 9.7	0.01	1.0e9	on	4.4E-12 to 2.2E-10	2	1
scalar_x0.02_kzz1e9	atomic	2.13	0.02	1.0e9	on	8.3E-11	1	1
scalar_x0.03_kzz1e9	atomic	2.13	0.03	1.0e9	on	4.4E-11	1	1
scalar_x0.05_kzz1e9	atomic	2.13	0.05	1.0e9	on	6.4E-11	1	1
scalar_x0.07_kzz1e9	atomic	2.13, 9.7	0.07	1.0e9	on	7.0E-12 to 5.1E-11	2	2
scalar_x0.10_kzz1e9	atomic	2.13, 9.7	0.10	1.0e9	on	4.4E-12 to 1.5E-11	2	2
scalar_x0.15_kzz1e9	atomic	2.13, 9.7	0.15	1.0e9	on	8.2E-12 to 9.1E-12	2	2
scalar_x0.20_kzz1e9	atomic	2.13, 9.7	0.20	1.0e9	on	5.5E-12 to 1.1E-11	2	2
scalar_x0.25_kzz1e9	atomic	2.13, 9.7	0.25	1.0e9	on	5.2E-12 to 1.7E-11	2	2
scalar_x0.30_kzz1e9	atomic	2.13, 9.7	0.30	1.0e9	on	3.2E-12 to 5.6E-12	2	2

group	wind chemistry	He/H	XUV	K_{zz}	diff.	du reached	cases	cert.
scalar_x0.33_kzz1e9	atomic	2.13, 9.7	0.33	1.0e9	on	2.8E-12 to 3.4E-12	2	2
photochem_kzzprofile	molecular	9	1	profile	on	1.4E-02	1	0
scalar_kzz1e9	molecular	0.083 to 9.7	1	1.0e9	on	9.3E-13 to 1.4E-02	4	2
scalar_wellmixed	molecular	0.083, 0.55	1	none	off	9.1E-13 to 6.6E-12	2	0
total							94	83

Besides these, the catalog holds 46 flux-closure iterates $k < NN$, two archival copies and one study directory. Nine of the iterates carry a certificate; the other 37 were written by runs that asked for no time integration, so they are states that were post-processed and never advanced, and they are not failures of a solve.

3. The models that are not certified

Eleven of the 94 models carry no certificate. Each was solved, or attempted, or written without being advanced at all, and the certificate of the generation its index names refuses it. The last column lists every entry of that certificate that stands above its tolerance, with the measure and the cell, in the certificate's own order.

case	wind chemistry	He/H	XUV	K_{zz}	diff.	du	$\ R\ $	the rows the certificate refuses on
diffusion_check/scalar_kzz0_HeH0.55	atomic	0.55	1	0.0	on	–	–	mass 1.541 at cell 1; momentum 7.593e-3 at cell 500; energy 1.583 at cell 5; elemental He/H 2.546e-4 at cell 500
diffusion_check/scalar_kzz1e10_HeH0.55	atomic	0.55	1	1.0e10	on	–	–	mass 1.541 at cell 1; momentum 2.466e-3 at cell 500; energy 1.907 at cell 5; elemental He/H 9.644e-5 at cell 500
diffusion_check/scalar_kzz1e8_HeH0.55	atomic	0.55	1	1.0e8	on	–	–	mass 1.542 at cell 1; momentum 5.150e-3 at cell 500; energy 1.952 at cell 5; elemental He/H 1.790e-4 at cell 500
diffusion_check/scalar_kzz1e9_HeH0.55	atomic	0.55	1	1.0e9	on	–	–	mass 1.542 at cell 1; momentum 3.719e-3 at cell 500; energy 1.917 at cell 5; elemental He/H 1.354e-4 at cell 500
photochem_x0.01_kzzprofile/HeH9.7	atomic	9.7	0.01	profile	on	9.356E-03	1.163E+00	mass 5.880e-1 at cell 11; momentum 7.774e-7 at cell 495; energy 1.949 at cell 7; elemental He/H 3.538e-2 at cell 389
scalar_x0.01_kzz1e9/HeH9.7	atomic	9.7	0.01	1.0e9	on	4.424E-12	9.978E-01	mass 9.806e-1 at cell 2; momentum 8.724e-6 at cell 1; energy 1.071 at cell 1
photochem_kzzprofile/HeH9	molecular	9	1	profile	on	1.406E-02	3.167E-01	mass 1.041 at cell 2; momentum 1.450e-3 at cell 1; energy 1.021 at cell 1; H ₂ carrier 1.000 at cell 432; elemental He/H 1.494e-4 at cell 217
scalar_kzz1e9/HeH0.083	molecular	0.083	1	1.0e9	on	4.879E-12	1.369E-08	H ₂ carrier 8.150e-4 at cell 499; elemental He/H 2.503e-4 at cell 280
scalar_kzz1e9/HeH0.55	molecular	0.55	1	1.0e9	on	1.408E-02	5.086E-02	energy 4.634e-1 at cell 257; H ₂ carrier 8.922e-3 at cell 261
scalar_wellmixed/HeH0.083	molecular	0.083	1	none	off	6.592E-12	2.701E-08	H ₂ carrier 7.371e-2 at cell 500
scalar_wellmixed/HeH0.55	molecular	0.55	1	none	off	9.086E-13	1.062E-08	energy 1.000 at cell 246; H ₂ carrier 2.363e-2 at cell 306

Read by mechanism, the eleven fall into three kinds.

- **The hydrodynamic Newton makes no progress** on the two models that remain at 0.01 of the fiducial XUV. The composition relaxes while the hydrodynamic rows stand still between passes. These are the models of the open solver item. The third model of that group, scalar_x0.01_kzz1e9/HeH2.13, left this table on 2026-09-21: an XUV continuation down the ladder 0.07, 0.05, 0.03, 0.02, 0.015, 0.01 reached a certified state of it, 17 to 19 passes a rung, every rung ending $\text{info} = 0$. It was reached by continuation and not by a better solver, and nothing establishes either way whether the two models still here have a stationary solution at all.
- **A mass, carrier or energy row refuses** the five molecular models, at cells 1 to 500.
- **Nothing was solved** in the four diffusion_check models: they were post-processed without a time integration, so they have no du and no $\|R\|$. Their certificates were measured all the same, and refuse on the hydrodynamic rows and the elemental row, which is what an unadvanced state is expected to do; they are not failures of a solve.

What the two numbers say. du is the fractional spread of the mass flux $\rho v r^2$ over the wind window, the quantity a marching run stops on, and $\|R\|$ is the stationary operator's residual norm. The recipe of this code, and the practice of the literature it follows, calls a wind converged at $du < 10^{-3}$. MEASURED on the models that carry no certificate,

naming the wind chemistry because the same short name occurs in both families: the molecular scalar_kzz1e9/He H0.083 reaches $du = 4.9 \times 10^{-12}$ with $\|R\| = 1.4 \times 10^{-8}$, the molecular scalar_wellmixed/HeH0.083 6.6×10^{-12} and 2.7×10^{-8} , the molecular scalar_wellmixed/HeH0.55 9.1×10^{-13} and 1.1×10^{-8} , and the atomic scalar_x0.01_kzz1e9/HeH9.7 4.4×10^{-12} with $\|R\| = 1.0$. By the flux-spread criterion these are converged winds, and three of them also sit at a residual norm of 10^{-8} .

What refuses them is this code's own row-by-row certification, which is stricter: it measures every equation of the inventory cell by cell and names each one that stands above its tolerance. Over the eleven the hydrodynamic mass row is the entry that appears most often, in seven of them, and the H_2 carrier balance in the wind in five. A state that meets the flux-spread criterion and fails one such row is a meaningful solution by the standard other work uses, and the table says which rows keep it from this one.

4. The models that are certified

Eighty-three models carry a certificate. The table lists each with the mass-loss rate and the He I 10830 equivalent width the catalog reports for it. Seven of these rows are new since 2026-09-20: the six rungs of the 2026-09-21 XUV continuation on the He/H = 2.13 branch at 0.015, 0.02, 0.03, 0.05 and 0.07 of the fiducial spectrum together with the 0.07 rung on the He/H = 9.7 branch, and the 0.01 rung on the He/H = 2.13 branch that the continuation certified. The two molecular rows at the foot certified on 2026-09-20, when every active equation of the inventory was measured on their states and found within tolerance.

case	wind chemistry	He/H	XUV	K_{zz}	diff.	du	$\ R\ $	$\log_{10} \dot{M}$	EW
photochem_x0.10_kzzprofile/HeH9.7	atomic	9.7	0.10	profile	on	4.973E-10	1.022E-07	6.840	0.2674
photochem_x0.15_kzzprofile/HeH9.7	atomic	9.7	0.15	profile	on	4.692E-12	6.210E-08	7.020	0.4822
photochem_x0.20_kzzprofile/HeH9.7	atomic	9.7	0.20	profile	on	2.002E-12	6.628E-08	7.160	0.6711
photochem_x0.25_kzzprofile/HeH9.7	atomic	9.7	0.25	profile	on	1.900E-12	6.245E-08	7.260	0.8429
photochem_x0.30_kzzprofile/HeH9.7	atomic	9.7	0.30	profile	on	1.261E-10	3.190E-08	7.350	1.0097
photochem_x0.33_kzzprofile/HeH9.7	atomic	9.7	0.33	profile	on	1.644E-10	3.457E-08	7.390	1.0979
scalarCNO_kzz1e9/HeH2.13	atomic	2.13	1	1.0e9	on	1.739E-11	1.352E-08	7.870	1.3721
scalar_gj699_wellmixed/HeH0.042	atomic	0.042	1	none	off	2.983E-13	1.341E-07	8.570	0.9091
scalar_gj699_wellmixed/HeH0.046	atomic	0.046	1	none	off	1.827E-13	9.505E-08	8.570	0.9857
scalar_gj699_wellmixed/HeH0.050	atomic	0.050	1	none	off	1.587E-11	1.414E-07	8.570	1.0608
scalar_gj699_wellmixed/HeH0.083	atomic	0.083	1	none	off	7.470E-13	1.320E-07	8.570	1.6271
scalar_gj699_wellmixed/HeH1	atomic	1	1	none	off	5.765E-13	8.372E-08	8.550	4.5250
scalar_gj699_wellmixed/HeH1000	atomic	1000	1	none	off	8.965E-13	7.482E-09	8.420	2.8725
scalar_kzz0/HeH0.55	atomic	0.55	1	0.0	on	1.234E-10	1.707E-08	7.840	0.0305
scalar_kzz0/HeH2.6	atomic	2.6	1	0.0	on	4.029E-11	1.344E-08	7.860	0.9549
scalar_kzz0/HeH3.0	atomic	3.0	1	0.0	on	3.264E-11	2.215E-08	7.870	1.0798
scalar_kzz0/HeH3.5	atomic	3.5	1	0.0	on	2.429E-11	1.458E-08	7.870	1.2175
scalar_kzz0/HeH3.7	atomic	3.7	1	0.0	on	4.900E-11	1.776E-08	7.870	1.2698
scalar_kzz0/HeH3.9	atomic	3.9	1	0.0	on	4.557E-11	2.821E-08	7.870	1.3168
scalar_kzz1e10/HeH0.55	atomic	0.55	1	1.0e10	on	1.963E-11	1.870E-08	7.860	0.6463
scalar_kzz1e10/HeH1.06	atomic	1.06	1	1.0e10	on	1.994E-11	2.280E-08	7.860	1.0456
scalar_kzz1e10/HeH1.15	atomic	1.15	1	1.0e10	on	2.240E-11	1.463E-08	7.860	1.1032
scalar_kzz1e10/HeH1.29	atomic	1.29	1	1.0e10	on	2.711E-11	1.336E-08	7.860	1.1867
scalar_kzz1e11/HeH0.55	atomic	0.55	1	1.0e11	on	1.769E-11	2.546E-08	7.860	0.8065
scalar_kzz1e11/HeH0.795	atomic	0.795	1	1.0e11	on	5.628E-11	1.651E-08	7.860	1.0302
scalar_kzz1e11/HeH0.865	atomic	0.865	1	1.0e11	on	7.733E-11	1.951E-08	7.860	1.0866
scalar_kzz1e11/HeH0.93	atomic	0.93	1	1.0e11	on	1.864E-10	2.097E-08	7.860	1.1364
scalar_kzz1e5/HeH2.6	atomic	2.6	1	1.0e5	on	3.749E-11	1.394E-08	7.870	0.9631
scalar_kzz1e5/HeH3.0	atomic	3.0	1	1.0e5	on	2.952E-11	1.531E-08	7.870	1.0880
scalar_kzz1e5/HeH3.35	atomic	3.35	1	1.0e5	on	2.488E-11	1.509E-08	7.870	1.1862
scalar_kzz1e5/HeH3.64	atomic	3.64	1	1.0e5	on	4.781E-11	1.764E-08	7.870	1.2627
scalar_kzz1e5/HeH3.93	atomic	3.93	1	1.0e5	on	4.160E-11	1.552E-08	7.870	1.3314
scalar_kzz1e6/HeH0.55	atomic	0.55	1	1.0e6	on	1.895E-11	2.047E-08	7.840	0.0591
scalar_kzz1e6/HeH2.4	atomic	2.4	1	1.0e6	on	2.785E-11	1.570E-08	7.860	0.9465
scalar_kzz1e6/HeH2.8	atomic	2.8	1	1.0e6	on	4.856E-11	1.407E-08	7.870	1.0804
scalar_kzz1e6/HeH3.19	atomic	3.19	1	1.0e6	on	4.002E-11	2.368E-08	7.870	1.1949
scalar_kzz1e6/HeH3.46	atomic	3.46	1	1.0e6	on	3.570E-11	1.742E-08	7.870	1.2669
scalar_kzz1e6/HeH3.74	atomic	3.74	1	1.0e6	on	6.781E-11	2.390E-08	7.870	1.3388
scalar_kzz1e7/HeH0.55	atomic	0.55	1	1.0e7	on	1.289E-10	2.116E-08	7.850	0.1585
scalar_kzz1e7/HeH2.70	atomic	2.70	1	1.0e7	on	2.053E-12	1.497E-08	7.870	1.1996
scalar_kzz1e7/HeH2.94	atomic	2.94	1	1.0e7	on	1.639E-12	1.420E-08	7.870	1.2724
scalar_kzz1e7/HeH3.18	atomic	3.18	1	1.0e7	on	4.895E-11	3.913E-08	7.870	1.3265
scalar_kzz1e8/HeH0.55	atomic	0.55	1	1.0e8	on	5.543E-11	2.155E-08	7.850	0.3089
scalar_kzz1e8/HeH2.05	atomic	2.05	1	1.0e8	on	7.949E-11	1.428E-08	7.870	1.1513
scalar_kzz1e8/HeH2.23	atomic	2.23	1	1.0e8	on	7.229E-11	1.895E-08	7.870	1.2206
scalar_kzz1e8/HeH2.41	atomic	2.41	1	1.0e8	on	5.482E-11	1.624E-08	7.870	1.2826
scalar_kzz1e9/HeH0.55	atomic	0.55	1	1.0e9	on	6.072E-11	1.518E-08	7.850	0.4793
scalar_kzz1e9/HeH1.50	atomic	1.50	1	1.0e9	on	2.944E-11	1.646E-08	7.870	1.0996
scalar_kzz1e9/HeH1.60	atomic	1.60	1	1.0e9	on	4.581E-11	1.313E-08	7.870	1.1511
scalar_kzz1e9/HeH1.70	atomic	1.70	1	1.0e9	on	3.779E-11	2.343E-08	7.870	1.1970
scalar_kzz1e9/HeH2.13	atomic	2.13	1	1.0e9	on	2.462E-11	1.700E-08	7.870	1.3749

case	wind chemistry	He/H	XUV	K_{zz}	diff.	du	$\ R\ $	$\log_{10} \dot{M}$	EW
scalar_kzz1e9/HeH4.0	atomic	4.0	1	1.0e9	on	9.341E-13	1.470E-08	7.870	1.8229
scalar_kzz1e9/HeH9.7	atomic	9.7	1	1.0e9	on	5.192E-11	2.009E-08	7.860	2.0592
scalar_wellmixed/HeH0.083	atomic	0.083	1	none	off	9.755E-12	2.543E-08	7.840	0.3797
scalar_wellmixed/HeH0.40	atomic	0.40	1	none	off	1.442E-12	2.641E-08	7.850	1.1011
scalar_wellmixed/HeH0.42	atomic	0.42	1	none	off	3.329E-12	1.823E-08	7.850	1.1324
scalar_wellmixed/HeH0.44	atomic	0.44	1	none	off	2.623E-12	1.723E-08	7.850	1.1626
scalar_wellmixed/HeH0.55	atomic	0.55	1	none	off	1.618E-12	1.929E-08	7.850	1.3118
scalar_wellmixed/HeH1	atomic	1	1	none	off	4.004E-12	1.485E-08	7.860	1.6959
scalar_wellmixed/HeH10	atomic	10	1	none	off	1.205E-10	1.281E-08	7.850	1.9347
scalar_wellmixed/HeH100	atomic	100	1	none	off	1.684E-12	1.075E-08	7.860	1.9644
scalar_wellmixed/HeH1000	atomic	1000	1	none	off	1.744E-12	1.567E-08	7.850	1.9197
scalar_x0.015_kzz1e9/HeH2.13	atomic	2.13	0.015	1.0e9	on	1.775E-10	9.233E-07	5.970	0.0000
scalar_x0.01_kzz1e9/HeH2.13	atomic	2.13	0.01	1.0e9	on	2.220E-10	3.154E-07	5.790	0.0000
scalar_x0.02_kzz1e9/HeH2.13	atomic	2.13	0.02	1.0e9	on	8.304E-11	1.096E-07	6.100	0.0000
scalar_x0.03_kzz1e9/HeH2.13	atomic	2.13	0.03	1.0e9	on	4.353E-11	1.169E-07	6.280	0.0000
scalar_x0.05_kzz1e9/HeH2.13	atomic	2.13	0.05	1.0e9	on	6.354E-11	2.969E-07	6.510	0.0000
scalar_x0.07_kzz1e9/HeH2.13	atomic	2.13	0.07	1.0e9	on	5.100E-11	1.976E-07	6.660	0.0000
scalar_x0.07_kzz1e9/HeH9.7	atomic	9.7	0.07	1.0e9	on	6.972E-12	9.931E-08	6.640	0.0445
scalar_x0.10_kzz1e9/HeH2.13	atomic	2.13	0.10	1.0e9	on	1.464E-11	3.886E-07	6.820	0.0001
scalar_x0.10_kzz1e9/HeH9.7	atomic	9.7	0.10	1.0e9	on	4.424E-12	4.362E-07	6.810	0.2353
scalar_x0.15_kzz1e9/HeH2.13	atomic	2.13	0.15	1.0e9	on	9.116E-12	1.239E-07	7.000	0.0003
scalar_x0.15_kzz1e9/HeH9.7	atomic	9.7	0.15	1.0e9	on	8.205E-12	1.162E-07	7.000	0.4514
scalar_x0.20_kzz1e9/HeH2.13	atomic	2.13	0.20	1.0e9	on	1.072E-11	1.165E-07	7.130	0.0022
scalar_x0.20_kzz1e9/HeH9.7	atomic	9.7	0.20	1.0e9	on	5.463E-12	8.109E-08	7.130	0.6294
scalar_x0.25_kzz1e9/HeH2.13	atomic	2.13	0.25	1.0e9	on	5.191E-12	1.806E-07	7.240	0.0700
scalar_x0.25_kzz1e9/HeH9.7	atomic	9.7	0.25	1.0e9	on	1.677E-11	1.044E-07	7.230	0.7913
scalar_x0.30_kzz1e9/HeH2.13	atomic	2.13	0.30	1.0e9	on	3.218E-12	6.850E-08	7.320	0.1894
scalar_x0.30_kzz1e9/HeH9.7	atomic	9.7	0.30	1.0e9	on	5.587E-12	6.702E-08	7.310	0.9395
scalar_x0.33_kzz1e9/HeH2.13	atomic	2.13	0.33	1.0e9	on	2.761E-12	4.566E-08	7.360	0.2555
scalar_x0.33_kzz1e9/HeH9.7	atomic	9.7	0.33	1.0e9	on	3.376E-12	9.900E-08	7.360	1.0231
scalar_kzz1e9/HeH2.13	molecular	2.13	1	1.0e9	on	9.272E-13	1.090E-08	7.900	1.5603
scalar_kzz1e9/HeH9.7	molecular	9.7	1	1.0e9	on	3.491E-11	2.580E-08	7.940	2.4061

5. What the flux-spread criterion misses, on one certified case

The question the columns above raise is what a state that meets $du < 10^{-3}$ but fails this code's certification differs from the certified stationary solution BY. Measured on the atomic scalar_kzz1e9/HeH2.13, which is certified, against a state of the same case stopped after one outer pass of the same route (docs/du_stop_vs_stationary_20260920.md):

	certified stationary	flux converged, refused
du , the run's own gate	2.46E-11	1.83E-12
$\ R\ $, re-evaluated	1.61E-08	4.97E-08
the three hydrodynamic rows	within	within
elemental transport He/H, $r \geq 1.2$	within	5.21E-05 above 1.0E-05 at $r = 1.695$
verdict	CERTIFIED	NOT CERTIFIED, one entry

Both clear the flux-spread criterion by four to eight decades and ONE row separates them. The two states differ by at most 1.9 per cent in any profile, and the difference sits in the composition and the temperature of the layer the He I 10830 line forms in (the metastable column's median radius is $6.4 R_p$, its 5 to 95 per cent range 1.7 to $16.7 R_p$), not in $\rho v r^2$, which agrees to 0.055 per cent everywhere. That is why the flux-spread criterion is blind to it.

It reaches a measured quantity. With the catalog's own selection the He I 10830 red-pair equivalent width moves from 1.374915 to 1.356726 % Å, a change of 1.32 per cent, and the red depth by 1.11 per cent, while $\log_{10} \dot{M}$ is 7.87 for both at the two decimals the code prints. For scale, the He/H rungs of this group stand 0.178 % Å apart in the equivalent width, so the shift is a tenth of the spacing the ladder measures a composition with.

Against the measurement it is smaller still. The observed red-pair equivalent width is 1.108 ± 0.030 % Å (Cherubim et al. 2026, READ from LHS1140b/MODELS.md), and the two states differ by 0.018 % Å, which is 0.61 of one standard deviation. For a comparison with the data the two states are the same model: the refused one would be read as the same answer. What the certification buys is not a different observable but the statement that the state solves the elemental transport equation everywhere, which is what lets a later measurement, or a ladder that resolves 0.178 % Å in composition, rest on it.

LHS 1140 b, atomic_scalar_gj1132_kzz1e9/HeH2.13: two flux-converged states

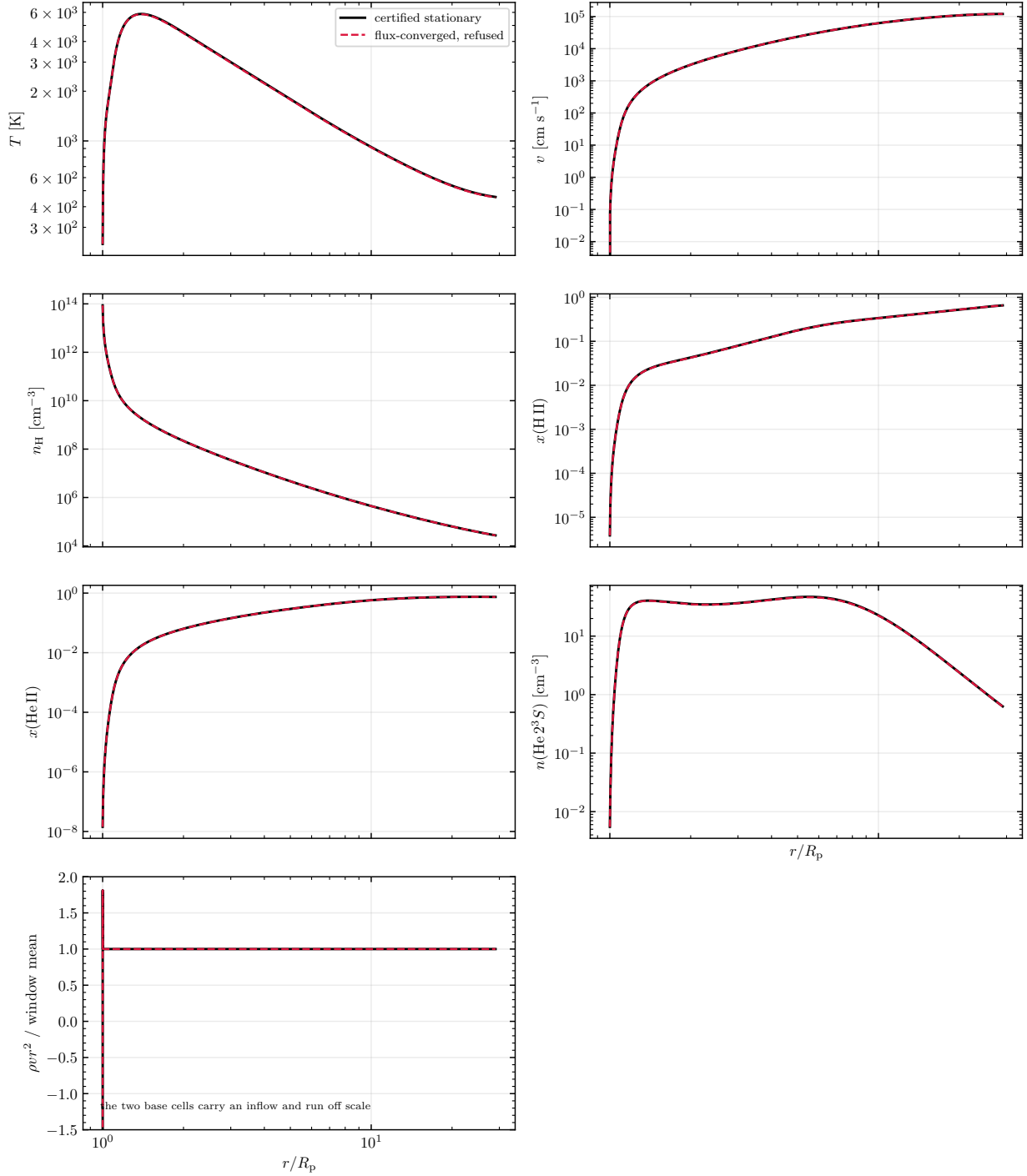


Figure 1: The certified stationary solution and the flux-converged refused state of the atomic scalar_gj1132_kzz1e9/HeH2.13, as functions of radius.

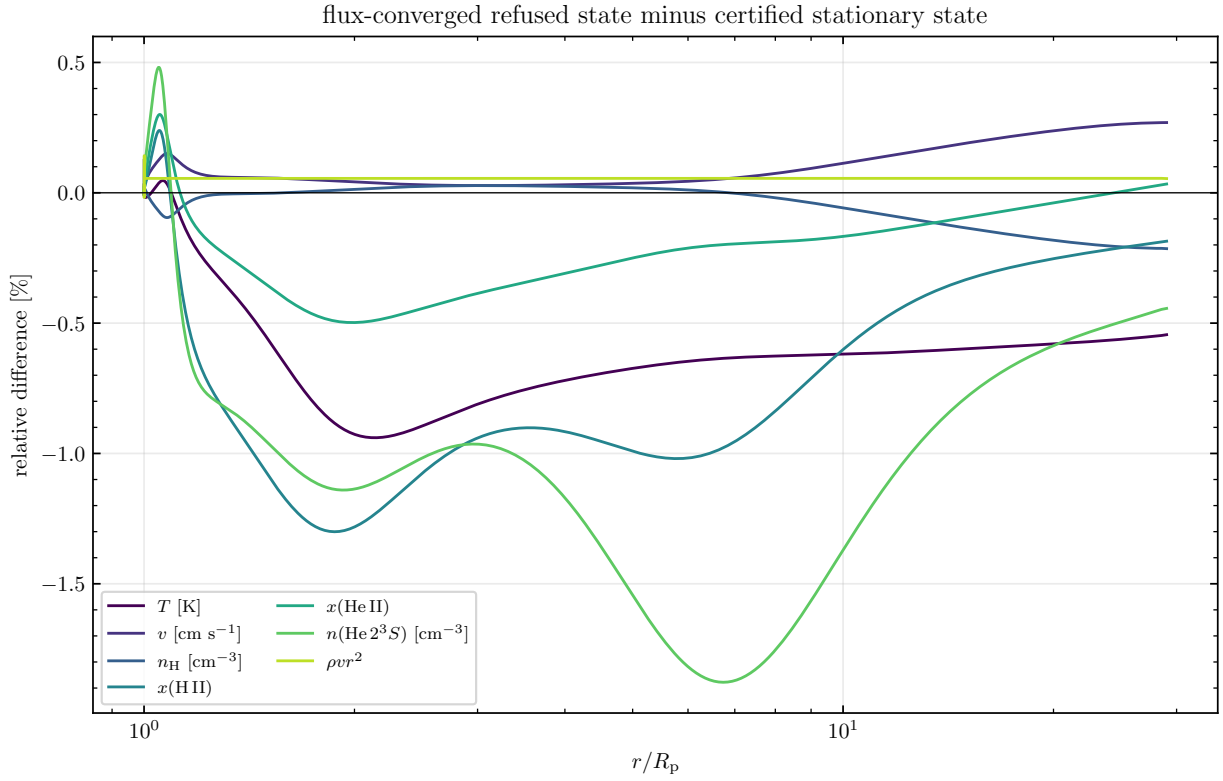


Figure 2: The same two states, as the relative difference of each quantity. The largest are the temperature at $r = 2.13 R_p$ (-0.94 per cent), $x(\text{He II})$ at 1.87 (-1.30) and $n(\text{He } 2^3S)$ at 6.74 (-1.88).

He I 10830 transmission, instrument convolved at $R = 68000$

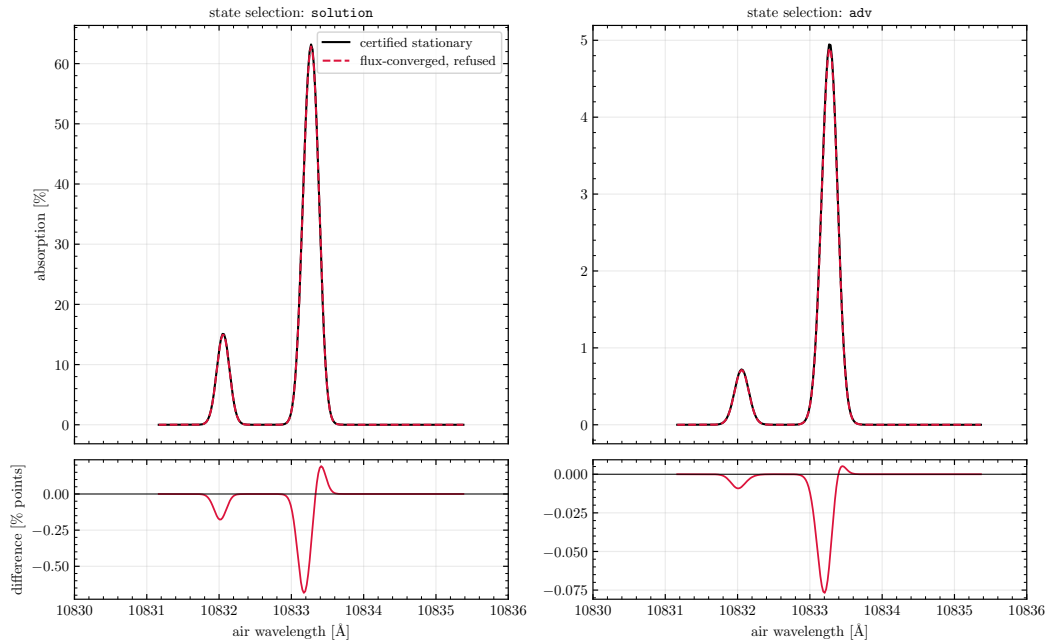


Figure 3: The He I 10830 transmission of the two states.